Communication Lab — Lab Sheet

Experiment: Simulation of Eye Diagram

Experiment No.: 8

Name: _____ **Roll No.:** _____

Date: _____ **Batch:** _____

1. Aim

1. To transmit an image by converting it to bits, mapping the bits to BPSK symbols, and pulse-shaping the symbols using an SRRC filter.
2. To model transmission over an AWGN channel and process the received waveform using a matched filter.
3. To generate and study the eye diagram of the received signal.
4. To observe how the eye diagram changes with roll-off factor β and signal-to-noise ratio (SNR).

2. Learning Objectives

At the end of this experiment, the student should be able to:

1. explain the purpose of an eye diagram in digital communication,
2. generate an eye diagram from a received waveform,
3. identify the best sampling instant from the eye opening,
4. relate eye closure to ISI, noise, jitter, and distortion,
5. study the effect of roll-off factor and SNR on signal quality.

3. Theory

An eye diagram is obtained by superimposing many short segments of a received digital waveform on the same plot. The resulting pattern resembles a human eye, hence the name *eye diagram*. It is a useful visual tool for assessing signal quality in a digital communication system.

In a practical receiver, the signal after matched filtering is sampled at suitable time instants to decide which symbol was transmitted. The eye diagram helps in identifying the most reliable sampling instant.

Important interpretations of an eye diagram are:

1. **Width of the eye opening:** indicates the time interval over which the signal can be sampled without significant error due to intersymbol interference (ISI).
2. **Maximum opening at the middle:** indicates the best sampling instant.
3. **Rate of closure of the eye:** indicates sensitivity to timing errors.
4. **Completely closed eye:** indicates severe ISI and poor signal quality.
5. **Asymmetric or squinted eye:** indicates distortion or non-ideal channel effects.

A sample eye diagram is shown in Figure 1.

In this experiment, an image is transmitted through the following stages:

Image → Bit stream → BPSK symbols → SRRC pulse shaping → AWGN channel → Matched filter → Eye diagram

Square-Root Raised Cosine (SRRC) Pulse

The SRRC pulse is commonly used for pulse shaping in digital communication systems. Its role is to control bandwidth and reduce ISI. A matched SRRC filter is used at the receiver.

A commonly used expression for the SRRC pulse is:

$$p(t) = \frac{4\beta}{\pi\sqrt{T_{\text{sym}}}} \cdot \frac{\cos((1+\beta)\pi t/T_{\text{sym}}) + \frac{T_{\text{sym}}}{4\beta t}\sin((1-\beta)\pi t/T_{\text{sym}})}{1-(4\beta t/T_{\text{sym}})^2}$$

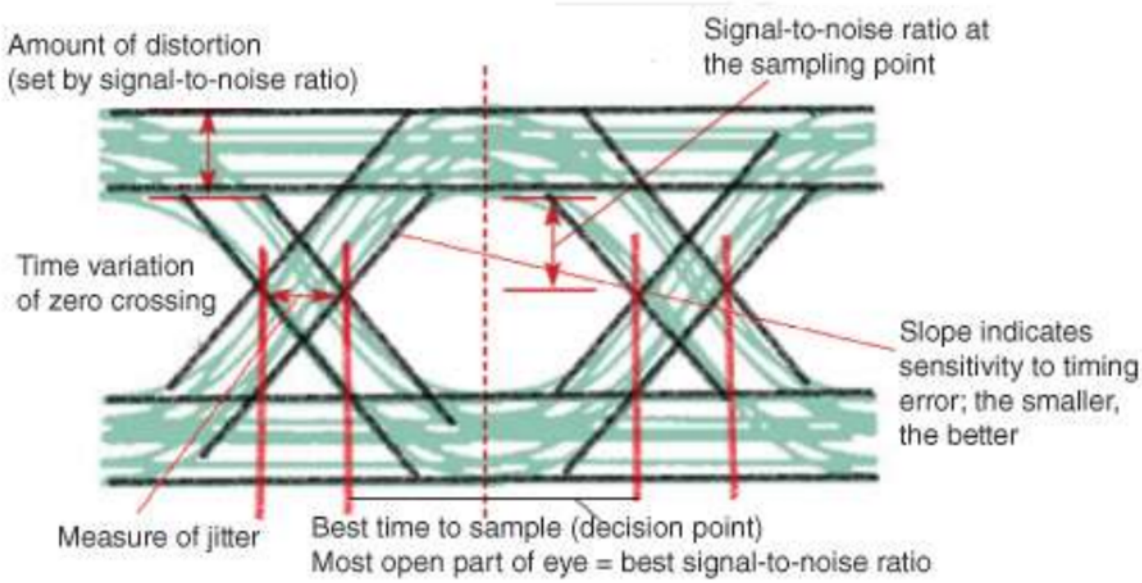

Figure 1: Eye Diagram

with suitable limiting values used at singular points.

In this experiment, the eye diagram is studied after matched filtering. The shape of the eye depends strongly on pulse shaping, noise, timing alignment, and the number of traces superimposed.

4. Pre-Lab Questions

Answer the following before coming to the lab:

1. What is an eye diagram? How is it obtained from a digital waveform?

2. What does the width of the eye opening represent?

3. Why is the middle of the eye opening considered the best sampling instant?

4. What is meant by jitter in the context of digital communication?

5. How do you expect the eye diagram to change when noise increases?

6. How do you expect the eye diagram to change when the roll-off factor β is varied?

5. Software / Tools Required

- Python 3.x
- NumPy
- Matplotlib
- OpenCV or imageio
- Jupyter Notebook / Google Colab / Python IDE

6. Parameters Used in the Experiment

Use the following default parameters unless otherwise instructed:

Table 1: Default simulation parameters

Parameter	**Value**
Image	`cameraman.png` (grayscale)
Modulation	BPSK
Upsampling factor	$L = 4$
Symbol duration	$T_{\text{sym}} = 1$
Pulse span	$N_{\text{sym}} = 8$ symbols
Matched filter	$g[n] = p[-n]$
Eye-diagram samples per trace	$nSamples = 3L$
Number of traces	$nTraces = 100$
Roll-off factor	vary β
SNR	vary as instructed

7. Procedure

1. Read the grayscale image `cameraman.png`.
2. Convert the 2D image array into a 1D array.
3. Convert each pixel value into binary using the same number of bits for all pixels.
4. Convert the binary strings into a serial bit stream.
5. Map the bits to BPSK symbols using:
$$0 \to -1, \qquad 1 \to +1$$
6. Plot the BPSK constellation.
7. Upsample the BPSK sequence by a factor of L.
8. Generate the SRRC pulse $p[n]$ using
$$t = -\frac{N_{\text{sym}}}{2} : \frac{1}{L} : \frac{N_{\text{sym}}}{2}$$
9. Convolve the upsampled sequence with the SRRC pulse to obtain the transmitted waveform.
10. Add AWGN corresponding to the chosen SNR.
11. Pass the received signal through the matched filter $g[n] = p[-n]$.
12. Compute the total delay introduced by the two FIR filters:
$$\delta = \frac{N-1}{2}, \qquad \text{total delay} = 2\delta$$
where N is the filter length.
13. Start sampling from the first valid sample and downsample every L-th sample.
14. Take $nSamples$ samples per trace and superimpose $nTraces$ such traces to generate the eye diagram.
15. Plot the eye diagram for a fixed β and two different SNR values.
16. Plot the eye diagram for a fixed SNR and different values of β.
17. Compare the eye diagrams and record your inferences.

8. Python Implementation Template

8.9 Eye Diagram Plot

```
nSamples = 3*L
nTraces = 100

plt.figure()
k = 0
for i in range(nTraces):
    trace = down[k:k+nSamples]
    if len(trace) == nSamples:
        plt.plot(np.real(trace))
    k += nSamples

plt.grid(True)
plt.title('Eye Diagram')
plt.xlabel('Sample index')
plt.ylabel('Amplitude')
plt.show()
```

8.10 Variation with SNR and Roll-off Factor

Repeat the eye-diagram plot for:

- fixed $\beta = 0.4$, different SNR values - 0dB, 4db 8dB,
- fixed SNR=4dB, different β values - 0.1, 0.5, 0.9.

9. Tasks to be Performed During the Lab

1. Generate the eye diagram for a fixed roll-off factor $\beta = 0.4$ and SNR =0dB, 4db 8dB.
2. Plot the eye diagram for three SNR values while keeping β fixed.
3. Plot the eye diagram for three different roll-off factors (0.1, 0.5, 0.9) while keeping SNR fixed to 4dB.
4. Compare the eye opening in each case.
5. Identify the best sampling region from the eye diagram.
6. Comment on the presence of ISI, timing uncertainty, and noise.

10. Observations

10.1 Effect of SNR on Eye Diagram

Table 2: Observation table for variation with SNR

SNR (dB)	**Roll-off factor β**	**Eye opening**	**Inference**
0	0.4		
4	0.4		
8	0.4		

10.2 Effect of Roll-off Factor on Eye Diagram

Table 3: Observation table for variation with β

Roll-off factor β	**SNR (dB)**	**Eye opening**	**Inference**
0.1	4		
0.5	4		
0.9	4		

11. Plots to be Included in the Record

Draw the following plots:

1. Eye diagram for SNR=4dB, $\beta = 0.1, 0.5, 0.9$ values

2. Eye diagram for $\beta = 0.4$, SNR = 0dB, 4dB, 8dB.

12. Result

Write the result of the experiment briefly.

13. Inference / Discussion

14. Viva / Post-Lab Questions

1. How does the eye diagram indicate the best sampling instant?

2. What happens to the eye opening when SNR decreases?

3. What happens to the eye diagram when the roll-off factor is changed?

4. What does a nearly closed eye indicate about the received signal?

Student's Signature: _____